An Adaptive Grouped-Heterogeneity Estimator for Cross-Sectional Data

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Setup

Consider the regression model

$$Y_i = \beta(W_i)' X_i + r_i + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid X_i, W_i] = 0$$

- Covariate $X_i \in \mathbb{R}^d$ may be continuous, and $W_i \in \{1, \dots, \mathcal{J}\}$ is discrete with large $\mathcal{J}$;
- Approximation error r_i vanishes as the sample size n grows.

Object of interest is the full collection $\beta_j := \beta(j), j = 1, \dots, \mathcal{J}$.

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Object of interest is the full collection $\beta_j := \beta(j), j = 1, \dots, \mathcal{J}$.

Problem: Least squares within each cell gives the naive estimator $\hat{\beta}_j$. Each cell averages $n/\mathcal{J}$ observations, so error accumulates over cells,

$$\sum_{j=1}^{\mathcal{J}} \|\hat{\beta}_j - \beta_j\|^2 \asymp_p d\mathcal{J}^2/n$$

Group structure: Different cells may share the same coefficient, so the β_j take only a few distinct values. Pooling cells within known groups would cut the loss to order $d\mathcal{J}/n$.

Example 1: Nonparametric Regression

Leading example is estimating $\mathbb{E}[Y_i \mid \tilde{W}_i, V_i]$, where $\tilde{W}_i$ collects several discrete covariates and V_i is an optional continuous covariate.

Let W_i index the cells of the saturated regression on $\tilde{W}_i$. Series regression within each cell gives

$$Y_i = \beta(W_i)' X_i + r_i + \varepsilon_i, \qquad X_i = p_d(V_i) \in \mathbb{R}^d$$

- Without V_i, $X_i = 1$ and $\beta(w) = \mathbb{E}[Y_i \mid W_i = w]$;
- k binary covariates give $\mathcal{J} = 2^k$ cells.

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Blandhol et al. (2026): Causal interpretability of 2SLS requires nonparametric estimate of the first-stage.

- Reanalyses have $(\mathcal{J}, n/\mathcal{J}) \approx (14, 220)$, $(99, 170)$, and $(66, 50)$.

Example 1: Existing Approaches

Saturated first stage meets the requirement of Blandhol et al. (2026) but is noisy.

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2. **Machine learning:** LASSO shrinks towards zero, not towards shared values. Restricted eigenvalue conditions may be suspect with discrete covariates. Theory for random forests and neural networks assumes continuous covariates;

3. **Clustering:** k-means needs the number of groups and is NP-hard as d or that number grows (Dasgupta, 2008, Aloise et al., 2009, Mahajan et al., 2012). Clustering can do worse than naive when groups are weakly separated or absent.

Example 2: Many Fixed Effects

Same framework covers models typically estimated by empirical Bayes. Simplest case is

$$Y_i = \theta(W_i) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid W_i] = 0$$

with $X_i = 1$ and $\beta(w) = \theta(w)$; Y_i may first be adjusted for common covariates.

- Teacher value-added (Kane et al., 2008): $\mathcal{J} = 10{,}040$ teachers, $n/\mathcal{J} \approx 150$;
- Medicare Advantage (Abaluck et al., 2021): $\mathcal{J} = 75{,}417$ plan products, $n/\mathcal{J} \approx 200$;
- Firm-level discrimination (Kline et al., 2022): $\mathcal{J} = 108$ firms, $n/\mathcal{J} \approx 770$.

Units with different labels may share the same effect.

- Teachers share training programs;
- Plans belong to insurer networks;
- Firms use common hiring practices.

Example 2: Empirical Bayes

Empirical Bayes reports posterior means under an estimated prior (Walters, 2024);

$$\hat{\beta}_j \mid \beta_j \sim N(\beta_j, \sigma_j^2), \qquad \beta_j \sim F, \qquad \hat{\beta}_j^{\mathrm{EB}} := \mathbb{E}_{\widehat{F}}\left[\beta_j \mid \hat{\beta}_j\right]$$

- Gaussian $\widehat{F}$ shrinks toward one center, so its rate stays naive under well-separated groups;
- Nonparametric maximum likelihood $\widehat{F}$ (NPMLE) attains the oracle rate up to logarithmic factors under strongly seperated groups (Saha and Guntuboyina, 2020, Soloff et al., 2025).

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However:

- Needs Gaussian errors, known covariances, fixed d
  - Chen et al. (2026) relax normality for $d = 1$
- Without group structure, NPMLE can perform worse than the naive estimator;
  - Simulated NPMLE loss is 1.09 to 1.16 times naive without groups, and 0.32 against 0.03 proposed with strong groups at $d = 5$.

Preview of Results

Proposed estimator pools similar cells, debiases, and recombines with the naive estimator using Stein's unbiased risk estimate (SURE).

1. Can the gains from pooling be obtained without knowing the number of groups or the assignment of cells to groups?
   - Answer: Yes. Under a fixed number of well-separated groups, the estimator attains the aggregate rate of an oracle that knows the partition, up to logarithmic factors.

2. Group structure is generally not verifiable. Is anything lost when it is weak or absent?
   - Answer: No. Aggregate quadratic loss is asymptotically no larger than that of the naive estimator, with no separation or grouping condition imposed.

Both established first in a Gaussian sequence model, then with estimated covariances, non-Gaussian errors, growing d, and nonzero approximation error.

Prior Literature

1. **Unobserved Groups.** Panels in Bonhomme and Manresa (2015), Su et al. (2016), Chetverikov and Manresa (forthcoming); clustering and fusion in Royer (2017), Stokell et al. (2021).
   - Proposed estimator uses closed-form weights and estimates no partition.

2. **Empirical Bayes.** Armstrong et al. (2022), Walters (2024), Chen (2026), Kwon (2026); NPMLE in Saha and Guntuboyina (2020), Soloff et al. (2025), Chen et al. (2026).
   - Proposed estimator needs no prior, is near-oracle under groups, and is asymptotically no worse than naive for any coefficients.

3. **Bilateral Filtering.** Tomasi and Manduchi (1998); limited statistical work, for images, in Arias-Castro et al. (2012); weights on unit-level estimates, without theory, in Chen et al. (2019).
   - Here cells are unordered, and the estimator adds debiasing, SURE, and large-sample theory.

Table of Contents

Gaussian Sequence Model

General Model

Monte Carlo

Empirical Application

Sequence Model

Suppose the naive estimates are exactly Gaussian,

$$\hat{\beta}_j = \beta_j + \sigma\xi_j, \qquad \xi_j \sim N(0, \Sigma_j), \qquad j = 1, \ldots, \mathcal{J}$$

independently across j, with Σ_j known, $\sigma^2 = \mathcal{J}/n \to 0$, d fixed, and no approximation error.

Benchmarks: Measure performance by aggregate quadratic loss;

- Naive: $\sum_{j=1}^{\mathcal{J}} \|\hat{\beta}_j - \beta_j\|^2 \asymp_p \mathcal{J}^2/n$;
- Group oracle, which pools within each known group $g(j) := \{k : \beta_k = \beta_j\}$: $\sum_{j=1}^{\mathcal{J}} \|\hat{\beta}_j^{\text{or}} - \beta_j\|^2 \asymp_p \mathcal{J}/n$.

Pooling over $\asymp \mathcal{J}$ cells per group improves the rate by a factor of $\mathcal{J}$.

Step 1: Exponential Weights

Average the naive estimates, weighting by closeness to $\hat{\beta}_j$ in a variance-adjusted metric;

$$\breve{\beta}_j := \sum_{k=1}^{\mathcal{J}} \omega_{jk} \hat{\beta}_k, \qquad \omega_{jk} := \frac{\exp\left(-\frac{1}{2\sigma^2}\|\hat{\beta}_j - \hat{\beta}_k\|_{\Omega_k}^2\right)}{\sum_{\ell=1}^{\mathcal{J}} \exp\left(-\frac{1}{2\sigma^2}\|\hat{\beta}_j - \hat{\beta}_\ell\|_{\Omega_\ell}^2\right)}, \qquad \Omega_\ell := \gamma(I_d - \gamma\Sigma_\ell)^{-1}, \quad \gamma > 0$$

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In the scalar case with $\Sigma_j = 1$, write $\Omega = \gamma/(1-\gamma)$;

$$\omega_{jk} \propto \begin{cases} \exp\left(-\Omega(\xi_j - \xi_k)^2/2\right), & k \in g(j) \\ \exp\left(-\Omega\left[|\beta_j - \beta_k|/\sigma + O_p(1)\right]^2/2\right), & k \notin g(j) \end{cases}$$

- Outside $g(j)$, weights vanish as $\sigma \to 0$;
- Within $g(j)$, weights stay random, and $\breve{\beta}_j$ averages $\asymp \mathcal{J}$ estimates of β_j.

Step 2: Debiasing

Problem: Weights depend on the noise in two ways;

- Each weight depends on every $\hat{\beta}_k$, so the terms of $\breve{\beta}_j$ are dependent;
- Given $\hat{\beta}_j$, the weights are nearly independent, but $\breve{\beta}_j$ is biased. Within $g(j)$, $\hat{\beta}_k$ gets more weight when ξ_k is close to ξ_j.

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Key Idea: Properties of the Gaussian distribution show that, under group structure,

$$\mathbb{E}\left[\breve{\beta}_j \mid \hat{\beta}_j\right] \approx (I_d - \Gamma_j)\beta_j + \Gamma_j \hat{\beta}_j, \qquad \Gamma_j := \gamma \sum_{k=1}^{\mathcal{J}} \omega_{jk} \Sigma_k$$

so the bias $\Gamma_j(\hat{\beta}_j - \beta_j)$ is of the order of the naive error. Solve for β_j to debias;

$$\tilde{\beta}_j := (I_d - \Gamma_j)^{-1}\left(\breve{\beta}_j - \Gamma_j \hat{\beta}_j\right), \qquad \mathbb{E}\left[\tilde{\beta}_j \mid \hat{\beta}_j\right] \approx \beta_j$$

Step 3: Recombination

Without well-separated groups, the debiased $\tilde{\beta}_j$ can do worse than naive. Recombine the two;

$$\check{\beta}_j(\alpha) := \alpha\tilde{\beta}_j + (1-\alpha)\hat{\beta}_j, \qquad \alpha \in [0,1]$$

Choose α to minimize Stein's unbiased risk estimate (SURE; Stein, 1981);

$$\widehat{\mathcal{L}}(\alpha) := \sum_{j=1}^{\mathcal{J}} \|\hat{\beta}_j - \check{\beta}_j(\alpha)\|^2 + 2\sigma^2 \sum_{j=1}^{\mathcal{J}} \operatorname{tr}\left(\Sigma_j \frac{\partial \check{\beta}_j(\alpha)}{\partial \hat{\beta}_j}\right) - \sigma^2 \sum_{j=1}^{\mathcal{J}} \operatorname{tr}(\Sigma_j)$$

- Unbiased for the risk $\mathbb{E}\sum_j \|\check{\beta}_j(\alpha) - \beta_j\|^2$ at each fixed α;
- Quadratic in α, and $\partial\tilde{\beta}_j/\partial\hat{\beta}_j$ has a closed form.

Proposed estimator is $\check{\beta}_j := \check{\beta}_j(\check{\alpha})$, with one $\check{\alpha} := \arg\min_{\alpha\in[0,1]} \widehat{\mathcal{L}}(\alpha)$ for all cells.

Stein's Identity | Closed-Form Jacobian

Sequence Model: Adaptivity

Theorem 1 (Adaptivity to Group Structure)

Suppose the sequence model and well-separated groups conditions hold, and $\mathcal{J}\log^3(\mathcal{J})/n \to 0$. *Then*

$$\sum_{j=1}^{\mathcal{J}} \|\tilde{\beta}_j - \beta_j\|^2 = O_p\Big(\frac{\mathcal{J}}{n}\Big), \qquad 1-\check{\alpha} = O_p\Big(\frac{\log^{3/2}\mathcal{J}}{\sqrt{\mathcal{J}}}\Big), \qquad \sum_{j=1}^{\mathcal{J}} \|\check{\beta}_j - \beta_j\|^2 = O_p\Big(\frac{\mathcal{J}\log^3(\mathcal{J})}{n}\Big).$$

Well-separated groups: Coefficients take at most G values, at least δ apart, each in $\asymp \mathcal{J}$ cells.

Sequence Model: Weak Domination

Theorem 2 (Weak Domination of the Naive Estimator)

Suppose the sequence model conditions hold. Then

$$\sum_{j=1}^{\mathcal{J}} \|\check{\beta}_j - \beta_j\|^2 \leq \big(1 + o_p(1)\big) \sum_{j=1}^{\mathcal{J}} \|\hat{\beta}_j - \beta_j\|^2.$$

SURE Oracle Inequality

Sequence Model: Weak Domination

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Interpretation:

- Separation is generally not verifiable, so the theorem puts no restriction on the coefficients;
- Bound is on aggregate loss, so some cells can still do worse than naive;
- Close but unequal coefficients may still be partly pooled.

Proof Idea. SURE tracks realized loss uniformly in α, with error small relative to the naive loss. Its minimizer $\check{\alpha}$ therefore does at least as well as $\alpha = 0$, the naive estimator.

SURE Oracle Inequality

Table of Contents

Gaussian Sequence Model

General Model

Monte Carlo

Empirical Application

General Model: Relaxing the Sequence Model

Return to the regression model and relax the sequence model in four directions;

1. Covariances unknown, replaced by heteroskedasticity-robust estimates;
2. Covariate dimension grows, $d \to \infty$;
3. Approximation error nonzero, $r_i \neq 0$;
4. Errors sub-Gaussian rather than Gaussian, independent and possibly heteroskedastic.

Estimator is unchanged, with estimated covariances in place of Σ_j.

Tuning: $\|\xi_j - \xi_k\|^2$ grows linearly in d, so set $\gamma = \kappa/d$;

- Within a group, weights remain random;
- Across groups, separation of order $\sqrt{d}$ still sends the weights to zero.

Tuning Parameter

General Model: Three Replacements

Weights and SURE depend on all $\mathcal{J}$ estimates, so per-cell central limit theorems do not suffice.

The sequence model argument uses three Gaussian tools;

1. **Exact Gaussian moments** motivate the debiasing matrix Γ_j.
   - Replaced by Lindeberg bounds, one Gaussian swap at a time (Lindeberg, 1922, Chatterjee, 2006).
2. **Stein's identity** makes SURE unbiased for aggregate risk.
   - Replaced by an approximate Stein identity from a leave-one-observation-out expansion.
3. **Gaussian Poincaré inequality** concentrates SURE around realized loss.
   - Replaced by an Efron–Stein argument over individual observations.

Lindeberg | Approximate Stein | Efron–Stein

General Model: Results

Both results assume the data generating process and approximation error conditions and $d \to \infty$.

Theorem 3 (Adaptivity to Group Structure)

Suppose groups are well separated and $\mathcal{J} d^2 \log^4(\mathcal{J} d)/n \to 0$. *Then*

$$\sum_{j=1}^{\mathcal{J}} \|\check{\beta}_j - \beta_j\|^2 = O_p\left(d\Big\{ \underbrace{\frac{\mathcal{J}}{n}}_{\text{oracle}} + \underbrace{\frac{\mathcal{J}^3}{n^2} \log^5(\mathcal{J} d)}_{\text{remainder}} \Big\} \right).$$

Remainder is negligible once $\mathcal{J}^2 \log^5(\mathcal{J} d)/n \to 0$. Either way, the bound is below the naive rate $d\mathcal{J}^2/n$.

General Model: Results

Both results assume the data generating process and approximation error conditions and $d \to \infty$.

Theorem 4 (Weak Domination of the Naive Estimator)

Suppose $\log^{12}(d)/\mathcal{J} + \mathcal{J}d^2 \log^6(\mathcal{J}d)/n \to 0$. *Then*

$$\sum_{j=1}^{\mathcal{J}} \|\check{\beta}_j - \beta_j\|^2 \leq \big(1 + o_p(1)\big) \sum_{j=1}^{\mathcal{J}} \|\hat{\beta}_j - \beta_j\|^2.$$

Table of Contents

Gaussian Sequence Model

General Model

Monte Carlo

Empirical Application

Monte Carlo: Design

Scalar cell means, so the naive estimates follow the sequence model;

$$Y_{ij} = \beta_j + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim N(0,1), \qquad \mathcal{J} = 160, \qquad m = n/\mathcal{J} \in \{20, 40, 80, 160\}$$

Coefficients

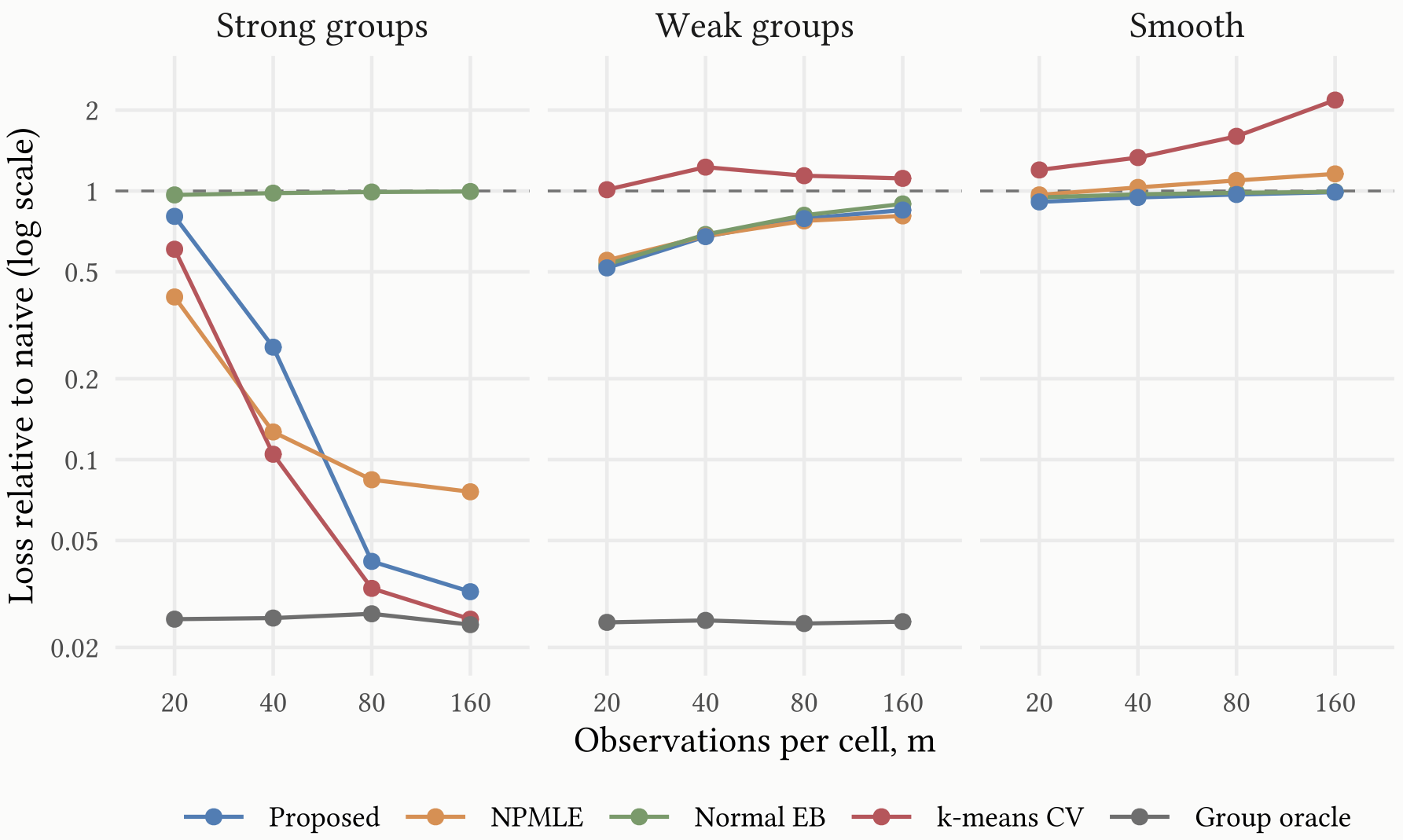
- Strong groups: $\beta_j \in \{-1.5, -0.5, 0.5, 1.5\}$;
- Weak groups: $\beta_j \in \{-0.3, -0.1, 0.1, 0.3\}$;
- Smooth: evenly spaced on $[-1.5, 1.5]$.

Comparisons

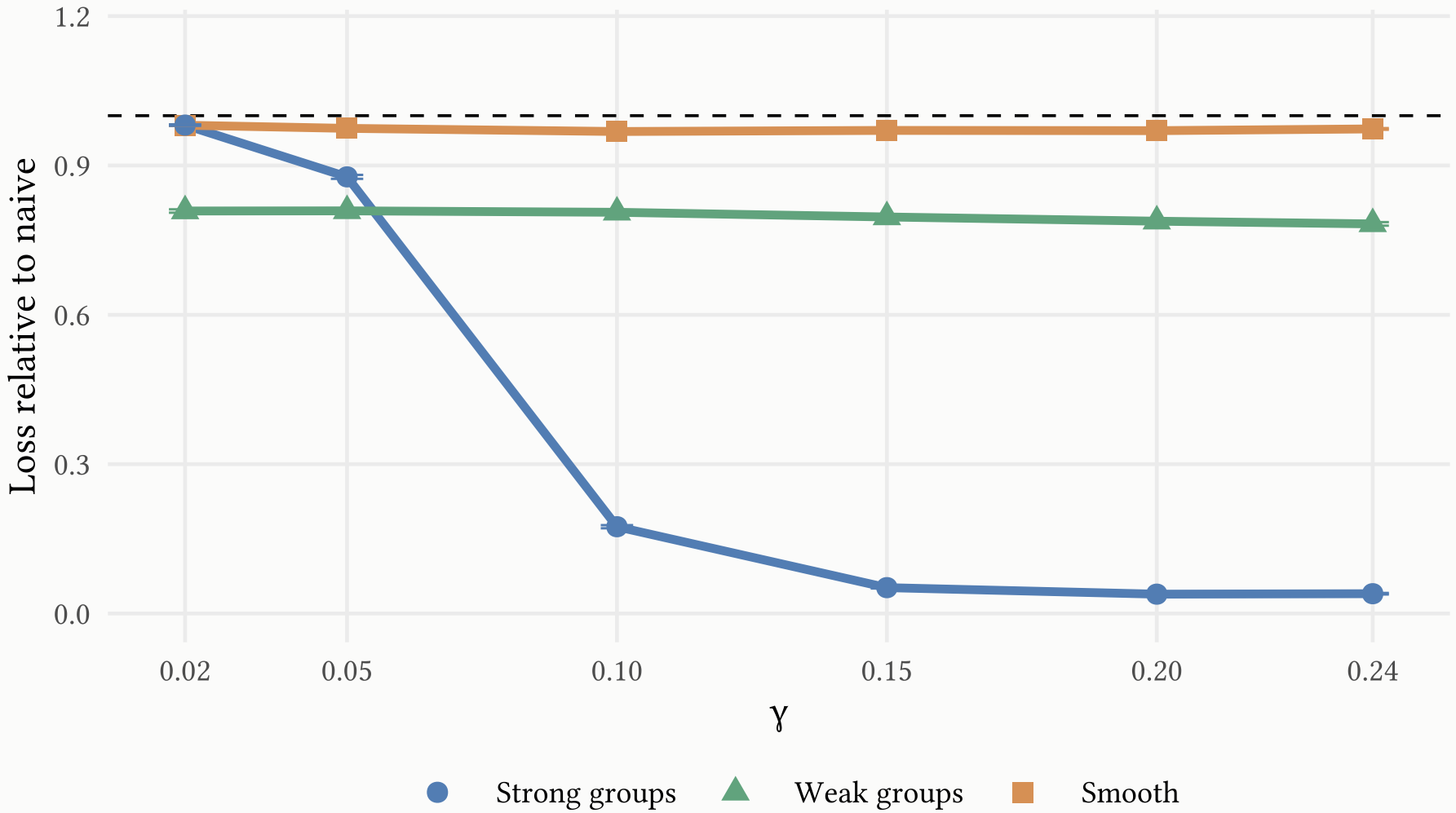
- NPMLE and Normal EB;
- Cross-validated k-means;
- Group oracle, infeasible.

Loss is aggregate squared error relative to naive in the same draw, averaged over 1,000 draws. Variances are estimated within cells, and $\gamma = 0.2$.

Monte Carlo: Scalar Designs



Monte Carlo: Sensitivity to γ



Monte Carlo: Five Dimensions, Uniform Errors

Same designs with $\beta_j \in \mathbb{R}^5$, Uniform$[-\sqrt{3}, \sqrt{3}]$ errors, $\gamma = 0.2/d$.

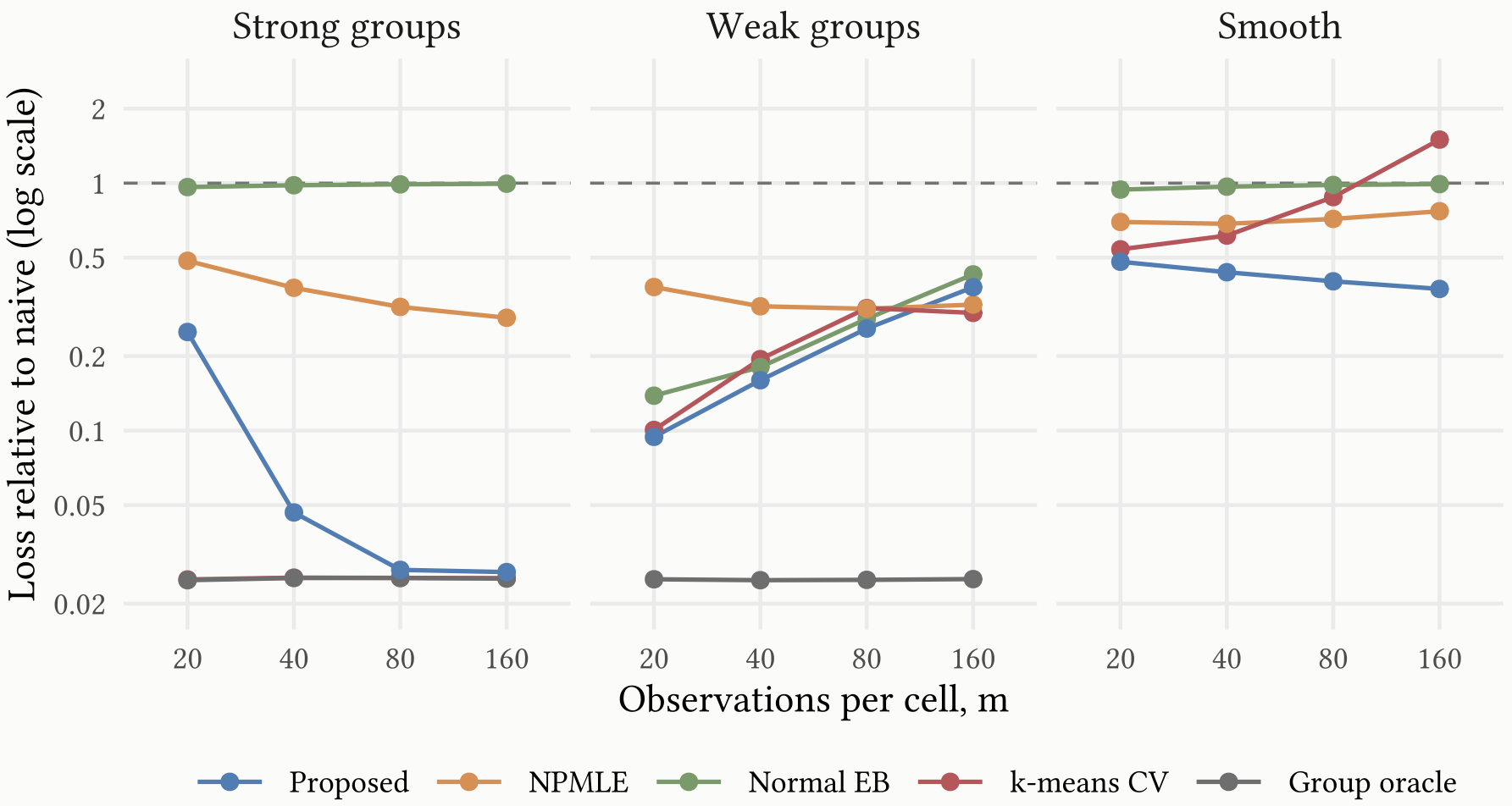


Table of Contents

Gaussian Sequence Model

General Model

Monte Carlo

Empirical Application

Angrist and Evans: Setting

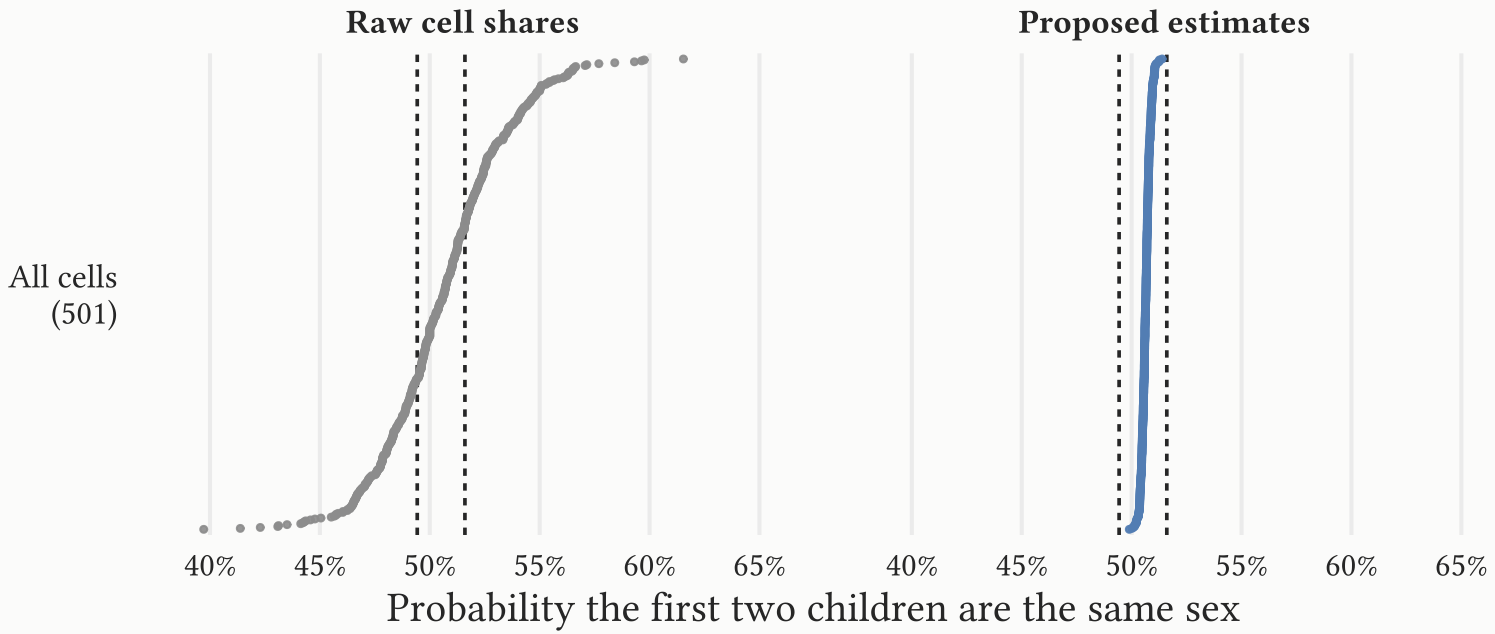
Angrist and Evans (1998): Does a third child reduce mothers' employment? Instrument Z_i for a third child is whether the first two children are the same sex.

- **Data:** 1980 Census mothers aged 21 to 35 with at least two children. Cells interact mother's age, age at first birth, race and ethnicity, and first child's sex, giving $\mathcal{J} = 501$;
- **Estimate:** Instrument propensity $p_j = \mathbb{E}[Z_i \mid W_i = j]$, the saturated first stage from Example 1;
- **References:** Sex at birth is nearly random, so p_j should be close to the full-sample shares, 49.4 percent after a first-born girl and 51.6 after a first-born boy;
- **Groups:** Reference points are close relative to cell-specific standard errors, covariates may still shift p_j slightly.

Estimator sees only the cell shares and their variances. We measure squared deviations from the references, relative to naive.

Sample

Angrist and Evans: Results

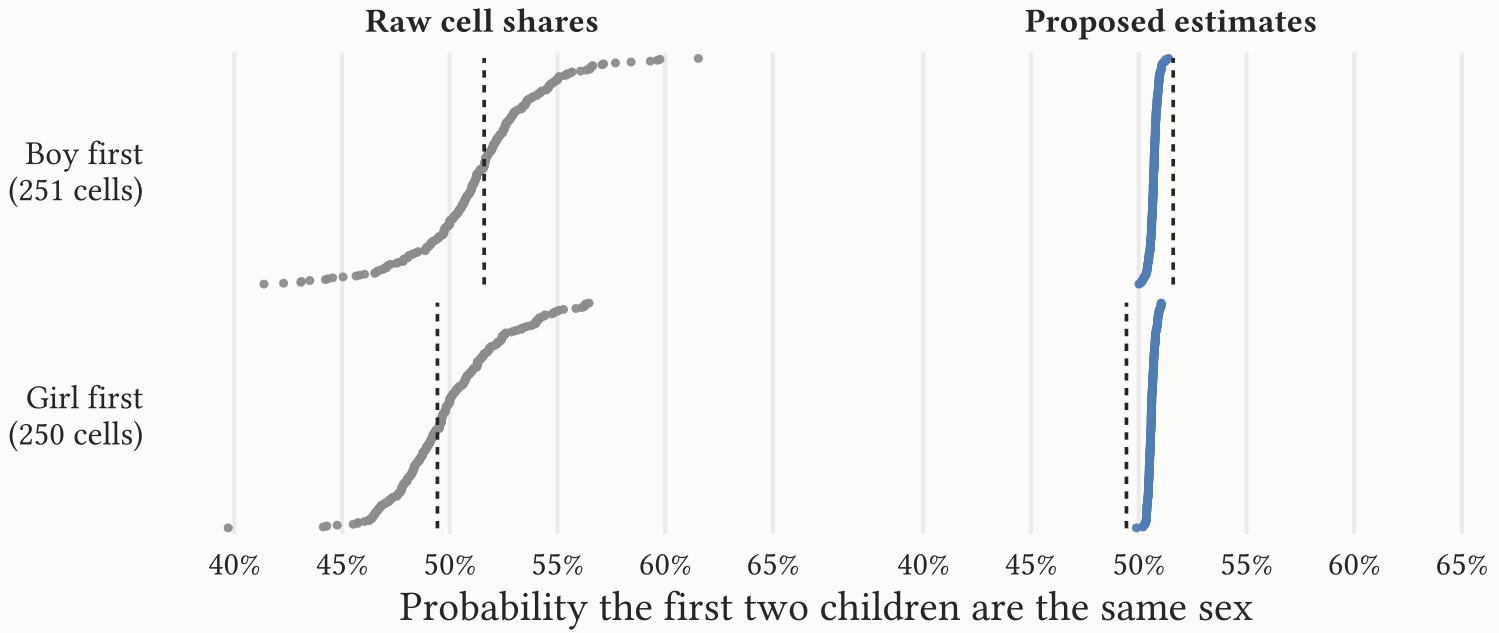


Rows are cells, sorted by raw share; dashed lines mark the two references.

Result: Reduces squared deviation from the references by 86 percent. SURE selects $\check{\alpha} = 0.936$.

Employment effect is essentially unchanged: −12.39 (2.57) points with the proposed propensity, −12.26 (2.57) with the saturated one.

Angrist and Evans: Results



Cells split by first-child sex; dashed lines mark each group's reference.

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Conclusion

This paper proposes an adaptive estimator for coefficients that vary with a discrete covariate of large support. This estimator

1. Pools similar cells, debiases, and recombines with the naive estimator using SURE, without estimating the groups;
2. Attains the oracle rate up to logarithmic factors under well-separated groups, and asymptotically has aggregate loss no larger than the naive estimator for any coefficients;
3. Extends to estimated covariances, non-Gaussian errors, growing covariate dimension, and approximation error, with an extra remainder in the rate.

`R` and `Stata` packages at github.com/mnavjeev/ewgroup and github.com/mnavjeev/ewgroup_stata.

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Thank you all very much

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Appendix: Technical Details

Formal Statement: Gaussian Sequence Model

Assumption (Gaussian Sequence Model)

Suppose that

(i) d is fixed, the coefficients $\beta_1, \dots, \beta_{\mathcal{J}}$ are nonrandom, and $\hat{\beta}_j = \beta_j + \sigma\xi_j$, $\xi_j \sim N(0, \Sigma_j)$, independently across j, where $\mathcal{J} \to \infty$ and $\sigma^2 = \mathcal{J}/n \to 0$;

(ii) The known, nonrandom matrices Σ_j are symmetric and, uniformly over j, $\underline{\lambda} \le \lambda_{\min}(\Sigma_j) \le \lambda_{\max}(\Sigma_j) \le \bar{\lambda}$;

(iii) γ is fixed and satisfies $0 < \gamma < (p\bar{\lambda})^{-1}$ for some $p > 4$.

Adaptivity | Weak Domination

Formal Statement: Well-Separated Groups

Assumption (Well-Separated Groups)

There are fixed constants $\delta > 0$, $\bar{B} < \infty$, and $G < \infty$ such that

(i) If $\beta_k \neq \beta_\ell$ then $\|\beta_k - \beta_\ell\| \geq \delta$, and $\max_{j \in [\mathcal{J}]} \|\beta_j\| \leq \bar{B}$;

(ii) The number of distinct values in $\{\beta_1, \ldots, \beta_{\mathcal{J}}\}$ is at most G;

(iii) For $g(j) = \{k \in [\mathcal{J}] : \beta_k = \beta_j\}$, $|g(j)| \asymp \mathcal{J}$ uniformly over j.

Imposed only for the adaptivity result. Since $\sigma \to 0$, the separation δ is large relative to the noise.

Back

SURE Oracle Inequality

Write $\mathcal{L}(\alpha) := \sum_{j=1}^{\mathcal{J}} \|\check{\beta}_j(\alpha) - \beta_j\|^2$ for the realized loss at mixing weight α, and define

$$R_{\mathcal{J}} := \frac{\mathcal{J}^{3/2} \log^{3/2}(\mathcal{J})}{n}$$

Proposition (SURE Oracle Inequality)

Suppose the sequence model conditions hold. For every fixed $\alpha \in [0,1]$, $\mathbb{E}\widehat{\mathcal{L}}(\alpha) = \mathbb{E}\mathcal{L}(\alpha)$. Moreover,

$$\sup_{\alpha \in [0,1]} \left|\widehat{\mathcal{L}}(\alpha) - \mathcal{L}(\alpha)\right| = O_p(R_{\mathcal{J}}) \quad \textit{and} \quad \mathcal{L}(\check{\alpha}) \leq \inf_{\alpha \in [0,1]} \mathcal{L}(\alpha) + O_p(R_{\mathcal{J}}).$$

- Stein's identity gives unbiasedness, and the Gaussian Poincaré inequality gives uniform concentration;
- Weak domination follows by taking $\alpha = 0$ in the second display, since $R_{\mathcal{J}}$ is small relative to the naive loss $\asymp \mathcal{J}^2/n$.

Back Gaussian Poincaré

Stein's Unbiased Risk Estimate

Lemma (Stein's Identity)

Let $\xi \sim N(0, \Sigma)$ in $\mathbb{R}^d$ and let $f : \mathbb{R}^d \to \mathbb{R}^d$ be almost differentiable with $\mathbb{E}\|Df(\xi)\| < \infty$, where $Df(x) := \partial f(x)/\partial x'$. Then

$$\mathbb{E}\langle f(\xi), \xi \rangle = \mathbb{E}\,\mathrm{tr}\left(\Sigma\, Df(\xi)\right).$$

Lemma (Stein's Unbiased Risk Estimate)

In the sequence model, let $\hat{\theta}_j = \hat{\theta}_j(\hat{\beta}_1, \ldots, \hat{\beta}_{\mathcal{J}})$ be almost differentiable in $\hat{\beta}_j$ with $\mathbb{E}\|\partial\hat{\theta}_j/\partial\hat{\beta}_j\| < \infty$, $j = 1, \ldots, \mathcal{J}$. Then

$$\mathrm{SURE}(\hat{\theta}) := \sum_{j=1}^{\mathcal{J}} \|\hat{\beta}_j - \hat{\theta}_j\|^2 + 2\sigma^2 \sum_{j=1}^{\mathcal{J}} \mathrm{tr}\left(\Sigma_j \frac{\partial \hat{\theta}_j}{\partial \hat{\beta}_j}\right) - \sigma^2 \sum_{j=1}^{\mathcal{J}} \mathrm{tr}(\Sigma_j)$$

satisfies $\mathbb{E}\,\mathrm{SURE}(\hat{\theta}) = \mathbb{E}\sum_{j=1}^{\mathcal{J}} \|\hat{\theta}_j - \beta_j\|^2$.

Idea. Expand $\|\hat{\theta}_j - \beta_j\|^2 = \|\hat{\theta}_j - \hat{\beta}_j\|^2 + 2\sigma\langle\hat{\theta}_j - \hat{\beta}_j, \xi_j\rangle + \sigma^2\|\xi_j\|^2$ and apply Stein's identity to ξ_j conditional on the other cells. Criterion $\widehat{\mathcal{L}}(\alpha)$ is $\mathrm{SURE}(\check{\beta}(\alpha))$.

Back

Closed-Form Jacobian

$$\frac{\partial \tilde{\beta}_j}{\partial \hat{\beta}_j} = I_d - (1 - \omega_{jj})H_j + H_j \sum_{\ell=1}^{\mathcal{J}} \omega_{j\ell}\left(\hat{\beta}_\ell - \hat{\beta}_j + \gamma \Sigma_\ell H_j N_j\right) c'_{j\ell}$$

$$H_j := (I_d - \Gamma_j)^{-1}, \qquad N_j := \sum_{\ell=1}^{\mathcal{J}} \omega_{j\ell}(\hat{\beta}_\ell - \hat{\beta}_j),$$

$$c_{j\ell} := g_{j\ell} - \sum_{k=1}^{\mathcal{J}} \omega_{jk} g_{jk}, \qquad g_{j\ell} := \sigma^{-2}\Omega_\ell(\hat{\beta}_\ell - \hat{\beta}_j)$$

Back

Gaussian Poincaré Inequality

In the sequence model, the Gaussian Poincaré inequality shows that SURE concentrates around realized loss.

Lemma (Gaussian Poincaré Inequality)

Let $\xi = (\xi_1, \dots, \xi_J)$ with $\xi_j \sim N(0, \Sigma_j)$ independent across j, and let $f : \mathbb{R}^{Jd} \to \mathbb{R}$ be almost differentiable with $\mathbb{E}\|\nabla f(\xi)\|^2 < \infty$. Then

$$\operatorname{Var}\big(f(\xi)\big) \le \sum_{j=1}^{J} \mathbb{E}\big[\nabla_j f(\xi)' \, \Sigma_j \, \nabla_j f(\xi)\big] \le \max_j \lambda_{\max}(\Sigma_j)\, \mathbb{E}\|\nabla f(\xi)\|^2,$$

where $\nabla_j f$ is the gradient with respect to ξ_j.

SURE Oracle Inequality · Efron–Stein

Formal Statement: Data Generating Process

Write

$$\mathcal{I}_j := \{i : W_i = j\}, \qquad G_j := \sum_{i \in \mathcal{I}_j} X_i X_i', \qquad S_j := \sum_{i \in \mathcal{I}_j} \tau_i^2 X_i X_i'$$

Assumption (Data Generating Process)

(i) Errors ε_i are independent with $\mathbb{E}[\varepsilon_i] = 0$, $\mathrm{Var}(\varepsilon_i) = \tau_i^2$, $\underline{\tau} \le \min_i \tau_i \le \max_i \tau_i \le \bar{\tau}$, and $\max_{i \le n} \|\varepsilon_i / \tau_i\|_{\psi_2} \le K_\varepsilon$;

(ii) Cells are balanced, $n_j \asymp n/\mathcal{J}$, with the scaled Gram matrix $n_j^{-1} G_j$, the scaled variance matrix $n_j^{-1} S_j$, and $\Sigma_j = \sigma^{-2} \mathrm{Var}(\hat{\beta}_j)$ all having eigenvalues bounded above and below;

(iii) Maximum within-cell leverage satisfies $h_n := \max_j \max_{i \in \mathcal{I}_j} X_i' G_j^{-1} X_i \lesssim d\mathcal{J}/n$;

(iv) The tuning parameter satisfies $\gamma = \kappa/d$ for a fixed $\kappa > 0$;

(v) $\mathcal{J} \to \infty$, $d\mathcal{J}/n \to 0$, and $\log(\mathcal{J}d) \lesssim d$.

- Part (i) replaces Gaussianity with sub-Gaussian, heteroskedastic errors;
- Part (v) keeps the total number of coefficients $d\mathcal{J}$ small relative to n.

Adaptivity | Weak Domination

Formal Statement: Well-Separated Groups (Growing d)

Assumption (Well-Separated Groups)

There are fixed constants $\delta > 0$, $\bar{B} < \infty$, and $G < \infty$ such that

(i) If $\beta_k \neq \beta_\ell$ then $\|\beta_k - \beta_\ell\| \geq \sqrt{d}\,\delta$, and $\max_{j \in [\mathcal{J}]} \|\beta_j\| \leq \sqrt{d}\,\bar{B}$;

(ii) The number of distinct values in $\{\beta_1, \ldots, \beta_{\mathcal{J}}\}$ is at most G;

(iii) $|g(j)| \asymp \mathcal{J}$ uniformly over j.

Separation is measured per coordinate on average. Noise has norm of order $\sigma\sqrt{d}$, so separation is still large relative to the noise.

Back

Formal Statement: Approximation Error

Assumption (Approximation Error)

For $\bar{r}_{2,j}^2 := n_j^{-1} \sum_{i \in \mathcal{I}_j} r_i^2$, the within-cell L_2 approximation error satisfies

$$\log^4(\mathcal{J}d) \max_{j \le \mathcal{J}} \bar{r}_{2,j} = o(n^{-1/2} + \mathcal{J}/n).$$

Limits how far approximation error moves the naive estimates and the residual-based covariance estimates. Holds under Hölder smoothness of the within-cell regression functions when d grows fast enough (Chen, 2007).

Adaptivity | Weak Domination

General Model: Tuning Parameter

Within a group, $\|\xi_j - \xi_k\|^2$ grows linearly in d. With a fixed γ, each other cell's weight would fall exponentially in d relative to cell j's own.

Set $\gamma = \kappa/d$ for a fixed $\kappa > 0$;

- Within a group, weights remain random;
- Across groups, separation of order $\sqrt{d}$ still sends the weights to zero.

Default $\gamma = 0.2/[d \max_\ell \lambda_{\max}(\Sigma_\ell)]$ has this scaling.

Back

Lindeberg Replacement

An interpolation bound replaces the Gaussian moment calculations behind Γ_j.

Lemma (Third-Order Lindeberg)

Let $X_1, \dots, X_N, Z_1, \dots, Z_N$ be mutually independent mean-zero random vectors in $\mathbb{R}^d$ with $\mathrm{Var}(X_i) = \mathrm{Var}(Z_i)$ for each i. Put $S = \sum_i X_i$ and $S^Z = \sum_i Z_i$. If $f : \mathbb{R}^d \to \mathbb{R}$ is three times continuously differentiable with $M_3(f) := \sup_x \sup_{\|u\|=\|v\|=\|w\|=1} |D^3 f(x)[u, v, w]| < \infty$, then

$$\left|\mathbb{E} f(S) - \mathbb{E} f(S^Z)\right| \le \frac{M_3(f)}{6} \sum_{i=1}^{N} \left(\mathbb{E}\|X_i\|^3 + \mathbb{E}\|Z_i\|^3\right).$$

Application: Shows that the weighted moments of the other cells' noise are close to their Gaussian values, so Γ_j still removes the leading bias.

Back

Approximate Stein

Stein's identity makes the SURE criterion unbiased but relies on Gaussian errors. Without normality, an approximate version holds for smooth functions of a sum of independent terms.

Lemma (Approximate Stein Identity)

Let $S = \sum_{i=1}^{N} X_i$ *with* $X_1, \dots, X_N$ *independent mean-zero random vectors in* $\mathbb{R}^d$ *with finite third moments, and put* $V := \mathrm{Var}(S)$. *If* $f : \mathbb{R}^d \to \mathbb{R}^d$ *is twice continuously differentiable with* $M_2(f) := \sup_x \sup_{\|u\|=\|v\|=1} \|D^2 f(x)[u, v]\| < \infty$, *then*

$$\left| \mathbb{E}\langle f(S), S\rangle - \mathbb{E}\,\mathrm{tr}\left(V\, Df(S)\right) \right| \le \frac{3}{2}\, M_2(f) \sum_{i=1}^{N} \mathbb{E}\|X_i\|^3.$$

Application: Shows that SURE is approximately unbiased for the risk in the general model.

Back

Efron–Stein

SURE must concentrate around realized loss uniformly over $\alpha \in [0, 1]$. Without normality, Efron–Stein replaces the Gaussian Poincaré inequality.

Lemma (Efron–Stein)

Let $X_1, \dots, X_N, X_1', \dots, X_N'$ be independent with X_i' distributed as X_i. For $X = (X_1, \dots, X_N)$, let $X^{(i)}$ replace X_i by X_i'. If $f(X) \in L^2$, then

$$\operatorname{Var}(f(X)) \leq \frac{1}{2} \sum_{i=1}^{N} \mathbb{E}\left[\left(f(X) - f(X^{(i)})\right)^2\right].$$

Application: Replacing one observation at a time shows that the gap between SURE and realized loss concentrates around its mean. Approximate Stein shows this mean is near zero.

Back Gaussian Poincaré

Angrist and Evans: Sample and References

Angrist and Evans (1998): Effect of a third child on whether the mother works for pay. Instrument Z_i is whether the first two children are the same sex.

- Cells are defined by mother's age, age at first birth, race and ethnicity, and sex of the first child;
- 1980 Census cells with at least 100 mothers give $\mathcal{J} = 501$ and $n = 365{,}491$;
- Saturated first stage estimates the instrument propensity $p_j = \mathbb{E}[Z_i \mid W_i = j]$ cell by cell.

References: Sex at birth is nearly random, so p_j should be close to the full-sample Census shares, 49.4 percent after a first-born girl and 51.6 after a first-born boy. Covariates may still shift p_j slightly, so the number of groups is not known.

Noise: Raw shares range from 39.7 to 61.5 percent, but only 6.2 percent of 95 percent intervals exclude the reference, against 5 percent by chance. Median standard error is 2.6 points.

Back